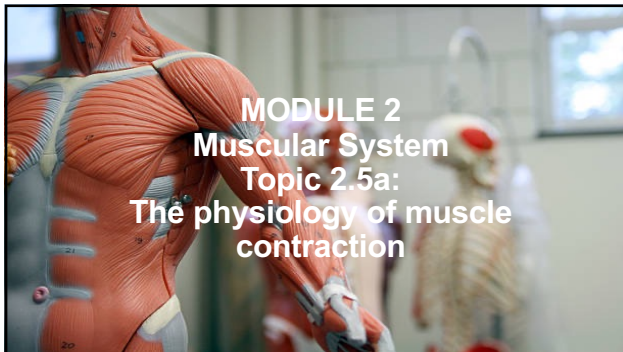




Learning Objectives

- ✓ Describe the microanatomy of a skeletal muscle cell and the significance of striations in muscle contraction
- ✓ Describe the features of the neuromuscular junction
- ✓ Describe the 4 stages of excitation-coupling from the arrival of action potential to the point when contraction begins
- ✓ Explain the stages of the cross – bridge cycle including the role of calcium and ATP.

Review Muscle Tissue

1. Skeletal - voluntary
2. Cardiac - involuntary
3. Smooth - involuntary

Skeletal muscle fibres

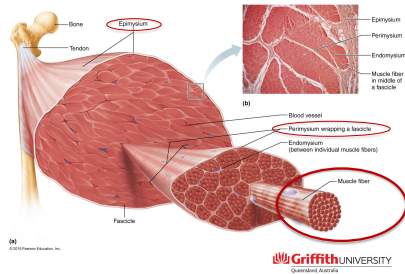
- Huge cells - long cylindrical
- Diameter: 10-100µm wide
- Length: 30cm long
- Multinucleated

CHARACTERISTIC	SKELETAL	CARDIAC	SMOOTH
Body location	Attached to bones or bones (via tendons) to skin	Walls of the heart	Unitary muscle in walls of hollow visceral organs (other than the heart); multi-unit muscle in iris/muscle, eye muscles, airway, large arteries
Cell shape and appearance	Single, very long, cylindrical, multinucleated cells with obvious striations	Branching chains of cells with or without striations	Single, spindle shaped, striated, no striations

Review: Gross Structure Skeletal Muscle

3 Connective tissue sheaths

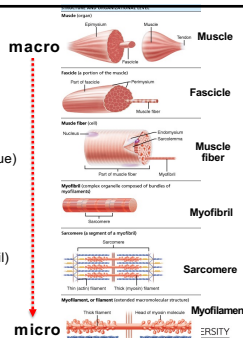
1. **Epimysium** ("outside")
 - surrounds whole muscle
2. **Perimysium** ("around")
 - bundles of muscle fibres (fascicles)
3. **Endomysium** ("within")
 - surrounds each muscle fibre



Microanatomy structure & organisation

Microscopic structures - terminology

1. **Muscle** (100-1000's muscle cells + connective tissue)
2. **Fascicle** (bundle of muscle cells)
3. **Muscle fiber** (elongated – multinucleate)
4. **Myofibril** (rods - bundles of myofilaments)
5. **Sarcomere** (contractile unit - segment of a myofibril)
6. **Myofilament** (contractile molecules)

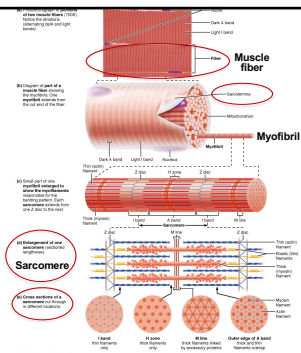


Microanatomy microscopic structures

Microscopic structures - terminology

muscle fibre → myofibril → myofilaments

- **Sarcomere** - The functional unit of a muscle fibre (2µm long)
- **Striations** - series of dark and light bands along a myofibril ('stripes')
- A muscle cell has 3 specialised structures:
 - i. **Myofibrils** (1-2µm diameter)
 - ii. **Sarcoplasmic reticulum (SR)**
 - iii. **T tubules**



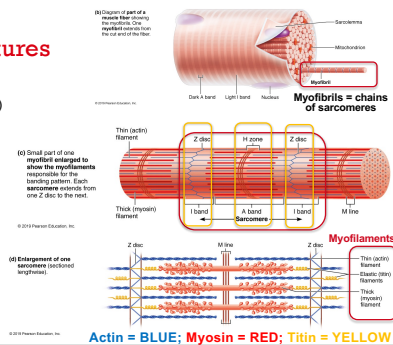
Microanatomy microscopic structures

i) Myofilaments (contractile molecules)

- Actin (thin)
- Myosin (thick)
- 'A' band = DARK Band
 - Actin + Myosin (overlap)
- 'I' band = LIGHT band
 - Actin + Titin (elastic filaments)

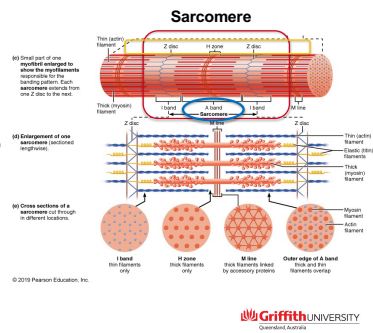
Sarcomere - functional unit

one A band + 1/2 I band each end
Z disc → Z disc



Microanatomy Sarcomeres

- H-zone: thick only →
M-line proteins holding myosin
- A band: DARK Band (thin & thick filaments)
 - overlap of actin & myosin
- I band: LIGHT Band (thin filaments)
 - actin anchored to Z disc
- Titin: elastic filament connects Z disc to thick filament - holds & recoils



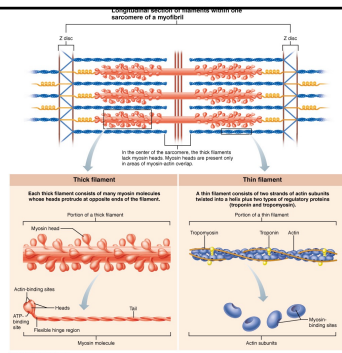
Microanatomy Myofilaments

Myosin

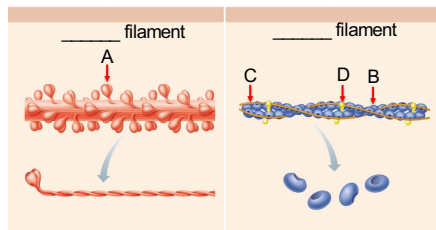
- 'head-like' structures attached to tail
- 'head' attaches to Actin to permit skeletal muscle contraction → forms **CROSS BRIDGE**
- myosin head has a bind site for ATP

Actin

- 2 strands (subunits) twisted into a helix
- contains a binding site for myosin (forms **cross bridge**)
- Tropomyosin**: covers (blocks) Actin binding site
- Troponin**: binds to Ca^{2+} (3 polypeptide complex)
 - Tn(I) inhibitory binds Actin
 - Tn(T) binds to tropomyosin
 - Tn(C) binds Ca^{2+}
- Troponin**: binds Calcium (from Sarcoplasmic reticulum) to expose myosin binding site on actin



Activity: 'Label & Learn'



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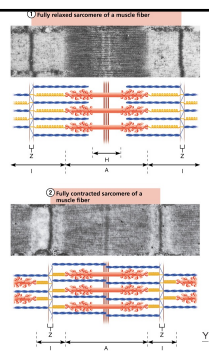
Sliding filament mechanism

At rest (no contraction):

- thin and thick filaments overlap only at the ends of the A band

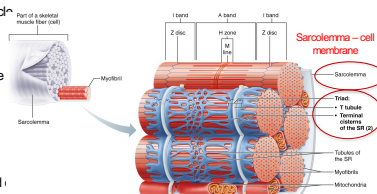
During contraction:

- thin filaments **slide** past the thick filaments so actin and myosin overlap to a greater degree
- H zones disappear**
- I bands shorten**
- Sarcomeres shorten (Z-Z)**
- Thick & thin filaments DO NOT change length during contraction** (Titin does)



Microanatomy Sarcoplasmic reticulum (SR)

- Sarcoplasm** - the 'cytoplasm' of a muscle cell (large amounts of glycogen [glucose] + myoglobin [O₂])
- The **sarcolemma** of the muscle cell, the cell membrane,
- T-tubules (light blue)** connect to the terminal cisternae of SR
- SR - Stores calcium (Ca²⁺)**
- Cell communication:** facilitates spread of **action potential** from sarcolemma to all myofibrils in muscle fibre



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What occurs BEFORE muscle fiber contraction?

Phase 1 - Neuromuscular Junction (NMJ)

- axon potential fires
- motor neuron contacts SM
- consists of an axon terminal and sarcolemma

Phase 2 - Excitation Contraction Coupling (ECC)

- AP down T tubules
- SR releases Ca^{2+}
- Ca^{2+} binds to troponin

Phase 3 - Cross bridge cycle

- contraction occurs (actin + myosin binding)

1. Neuromuscular junction

2. Excitation-contraction coupling

3. Cross-bridge cycle

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Phase 1 Neuromuscular Junction (NMJ)

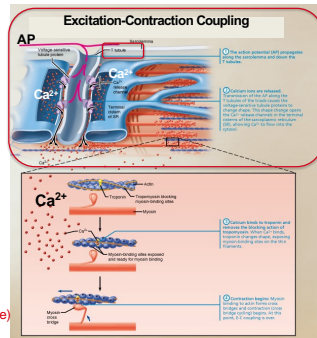
- The motor end plate
- Na^+ & K^+ chemical gates opened by ACh → local depolarisation
- Na^+ voltage gates open: wave of depolarisation spreads along sarcolemma
- generates action potential 1-2msec
- muscle contraction 20-200msec

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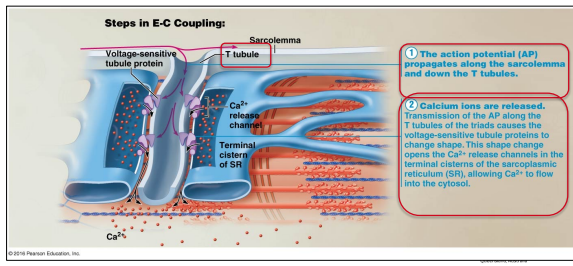
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Phase 2 Excitation-Contraction Coupling

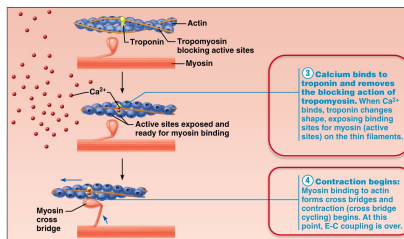
1. AP propagates along sarcolemma → down T-tubule (T= transverse)
2. Voltage change → triggers Ca^{2+} release from terminal cisternae of SR into cytosol
3. Ca^{2+} bind to troponin removing blocking action of tropomyosin
 - Myosin binding sites exposed
4. Events of sliding filament mechanism commence → myosin binds to actin (cross bridge)

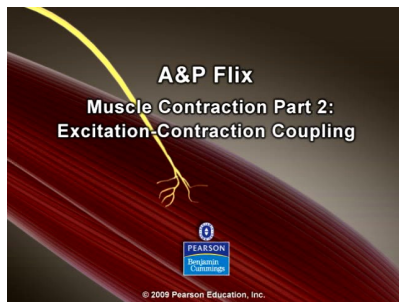


Phase 2 Excitation-Contraction Coupling



Phase 2 Excitation-Contraction Coupling



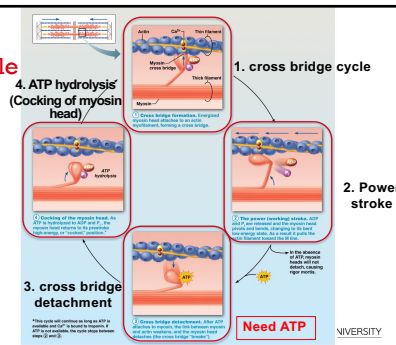


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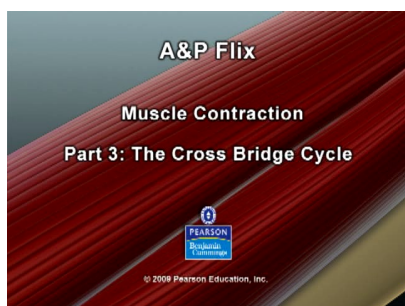
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Phase 3 Cross-bridge cycle

- 1. Cross-bridge formation:** energised myosin head attaches to active actin binding site
- 2. Power stroke** = ADP and P_i (inorganic Phosphate) released, myosin head pivots and bends, pulling actin filament along
- 3. Cross-bridge detachment** : after ATP attaches to myosin head (link between actin and myosin weakens)
- 4. ATP hydrolyses** = ADP + P_i , myosin head cocks into high-energy position



NIVERSITY



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Activity: Review & Learn

1. What causes the striations that characterise skeletal muscle?
2. Name the functional contractile unit of a muscle cell.
3. Name the two contractile myofilaments specially arranged in each myofibril.
4. Which myofilament has a binding site for calcium?
5. When a muscle fibre contracts the banding pattern changes. Describe this.
6. List the 3 stages of **excitation-contraction coupling** **prior** to the **cross-bridge cycle**.
7. What event causes the 'power stroke' in the cross-bridge cycle?
8. What two events are necessary for relaxation of the muscle fibre to occur?

Summary of key concepts: Muscular System 2.5a

- **Skeletal muscle** contains striations = vertically arranged light and dark bands along a muscle fibre.
- Dark bands = overlapping thick and thin myofilaments.
- The **sarcomere** = functional contractile unit of a muscle cell.
- Chains of sarcomeres exist in the myofibrils (a bundle of specially arranged myofilaments).
- When a muscle fibre contracts the banding pattern changes.
- The steps of **excitation-contraction** coupling includes:
 - spread of AP along sarcolemma to T-tubules
 - release of Ca from SR
 - binding of Ca to troponin and exposure of active site
- This is followed by **cross bridge cycle** = contraction of the muscle fibre
- ATP destabilises the cross bridges and pumps Ca^{2+} back into the SR for muscle relaxation to occur.
